

MUTSU, Japan

WMO: 475760

Lat: 41.2833N

Lon: 141.2114E

Elev: 5

StdP: 101.27

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.0	-6.5	-12.9	1.2	-5.5	-11.5	1.4	-4.3	9.1	0.5	8.1	0.2	1.9	290	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.5	29.2	24.0	27.6	23.1	26.1	22.2	24.7	27.9	23.9	26.7	23.1	25.5	3.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.7	18.5	26.7	23.0	17.7	26.0	22.2	16.9	25.1	75.2	28.1	72.1	26.9	68.9	25.6	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.1	6.1	-12.1	31.7	2.4	1.5	-13.8	32.8	-15.2	33.7	-16.5	34.5	-18.3	35.6		
			-12.4	25.8	2.4	0.8	-14.1	26.4	-15.5	26.9	-16.8	27.4	-18.5	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-1.6	-1.2	2.0	7.6	12.7	16.3	20.5	22.2	18.8	12.5	6.5	1.0
	DBStd	8.76	2.71	2.86	3.04	3.25	2.99	2.80	3.10	2.79	2.93	3.04	3.53	2.95
	HDD10.0	1421	361	313	247	85	9	0	0	0	0	11	115	279
	HDD18.3	3361	619	547	505	321	174	73	15	3	29	182	356	537
	CDD10.0	1363	0	0	0	14	94	189	326	379	264	88	9	0
	CDD18.3	262	0	0	0	0	1	12	82	123	43	1	0	0
	CDH23.3	1249	0	0	0	1	11	41	373	658	164	1	0	0
	CDH26.7	234	0	0	0	0	1	2	67	143	22	0	0	0
Wind	WSAvg	2.7	2.7	2.8	3.1	3.1	2.9	2.5	2.3	2.2	2.3	2.5	2.8	2.8
Precipitation	PrecAvg	1324	98	80	76	77	95	89	133	154	167	112	121	109
	PrecMax	1671	208	151	118	166	168	242	360	303	320	220	226	182
	PrecMin	979	46	34	35	11	31	18	22	48	37	43	57	39
	PrecStd	158	38	34	24	34	36	48	69	65	71	47	50	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	8.0	14.1	20.8	24.8	26.5	30.5	31.6	29.0	22.5	17.8	11.6
		MCWB	4.5	5.0	8.7	12.2	15.8	19.4	24.3	25.1	23.4	17.6	13.7	9.3
	2%	DB	5.0	6.0	11.1	17.5	22.3	24.5	28.6	29.7	26.8	20.7	15.7	9.3
		MCWB	2.8	3.4	7.0	11.0	14.8	18.7	23.6	24.5	22.5	16.5	12.3	7.0
	5%	DB	3.7	4.8	8.9	15.2	20.3	22.9	26.9	28.3	25.1	19.5	14.2	7.7
		MCWB	1.6	2.3	5.6	9.6	14.0	18.0	22.7	23.8	21.1	15.7	11.2	5.5
	10%	DB	2.7	3.6	7.2	13.3	18.3	21.4	25.4	26.8	23.8	18.3	12.8	6.0
		MCWB	0.6	1.3	4.2	8.7	13.3	17.5	22.0	23.0	20.2	14.9	10.2	3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	6.0	9.4	13.6	17.5	21.1	25.1	26.0	24.6	19.3	14.7	9.8
		MCDB	6.2	7.3	12.5	18.2	22.2	24.2	29.2	30.3	27.6	21.0	16.4	11.5
	2%	WB	3.2	3.8	7.6	12.0	16.1	19.8	24.2	25.1	23.3	17.6	13.1	7.6
		MCDB	4.5	5.5	10.4	15.6	20.0	23.1	27.4	28.5	25.8	19.7	15.1	9.0
	5%	WB	2.0	2.7	6.1	10.6	14.9	18.9	23.4	24.4	22.1	16.6	11.8	5.7
		MCDB	3.4	4.4	8.5	14.1	19.0	21.9	26.2	27.3	24.3	18.7	13.7	7.4
	10%	WB	0.9	1.5	4.6	9.3	13.8	17.9	22.5	23.7	21.0	15.5	10.5	4.1
		MCDB	2.3	3.2	6.9	12.5	17.4	20.7	24.9	26.1	23.0	17.7	12.4	5.9
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.3	7.3	8.8	8.8	7.7	6.5	6.5	8.0	9.4	7.8	6.0
		MCDBR	6.9	7.7	10.1	12.3	12.7	10.4	8.9	8.7	8.9	10.3	10.5	8.5
		MCWBR	5.8	6.0	7.1	6.9	6.5	5.6	4.6	4.4	5.2	6.9	7.9	7.2
	5% WB	MCDBR	6.7	7.6	9.5	10.8	11.0	9.0	8.0	7.8	7.7	8.6	9.6	8.4
		MCWBR	6.0	6.5	7.3	7.0	6.5	5.4	4.4	4.2	4.8	6.4	8.1	7.6
Clear-Sky Solar Irradiance	taub	0.365	0.409	0.481	0.535	0.551	0.524	0.523	0.492	0.434	0.413	0.397	0.380	
	taud	2.116	1.999	1.874	1.804	1.853	2.006	2.082	2.181	2.298	2.219	2.153	2.101	
	Ebn at Noon	747	768	754	746	747	769	765	777	800	763	706	687	
	Edh at Noon	108	141	180	207	202	174	160	141	117	113	104	102	
All-Sky Solar Radiation	RadAvg	1.43	2.20	3.36	4.50	4.93	4.84	4.28	4.12	3.65	2.79	1.68	1.19	
	RadStd	0.08	0.19	0.23	0.46	0.50	0.39	0.46	0.44	0.32	0.22	0.14	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page